Statistical Modelling and Hypothesis

Course: DataAnalytics/Statistics

Project: Customer Engagement andPurchasingBehaviour.

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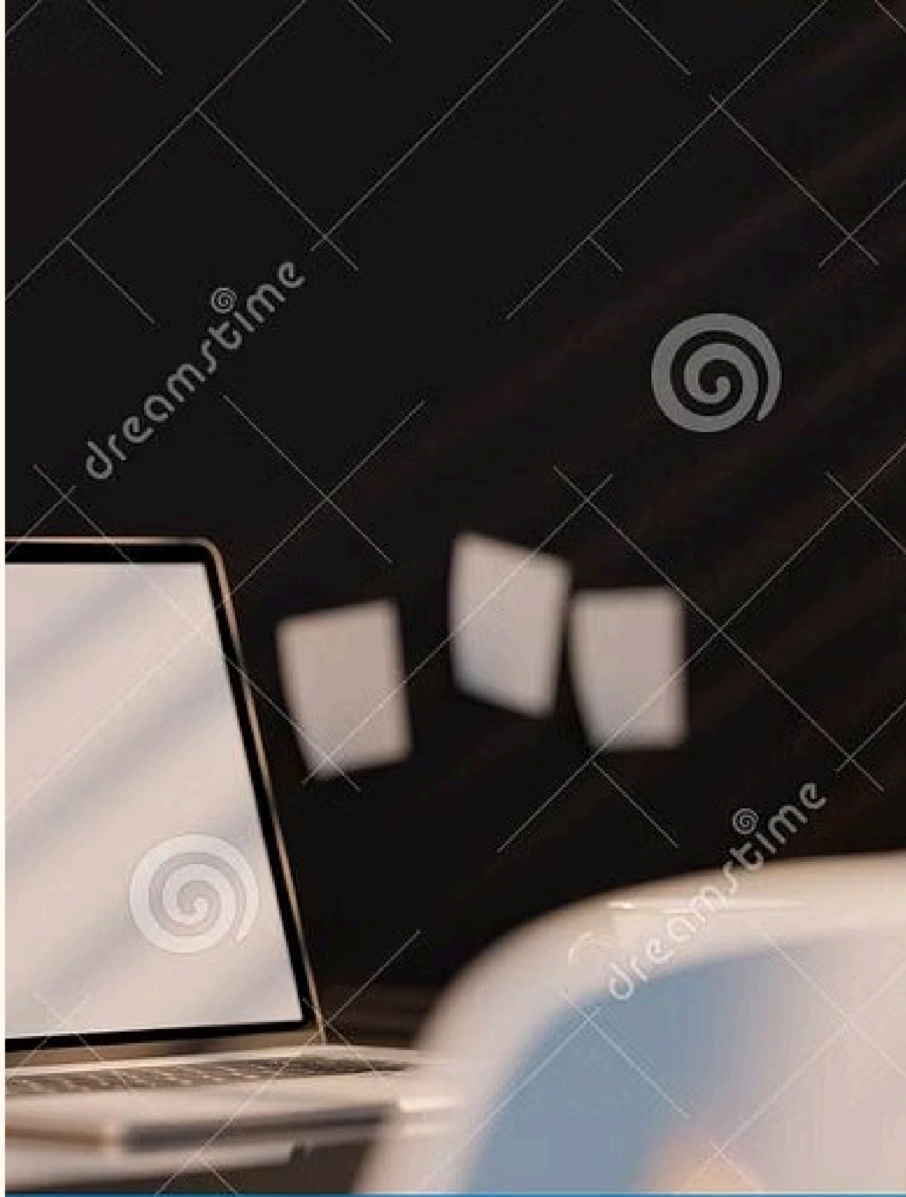
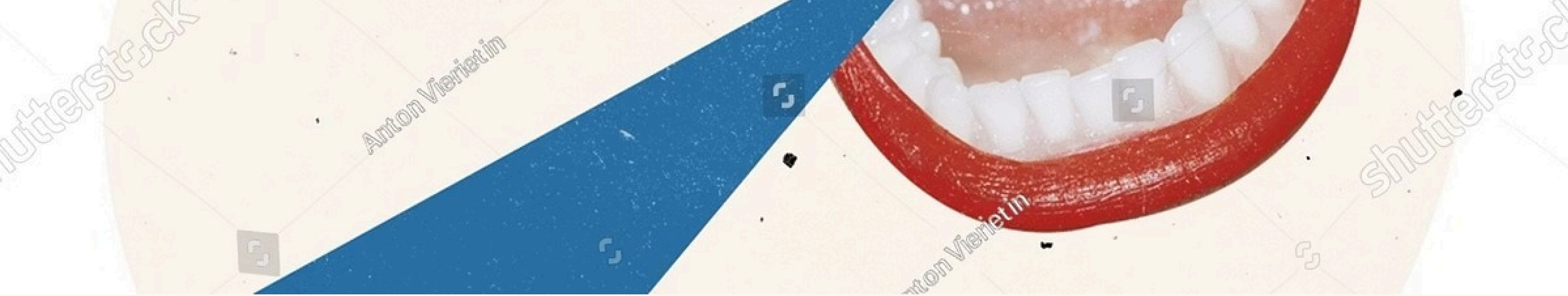




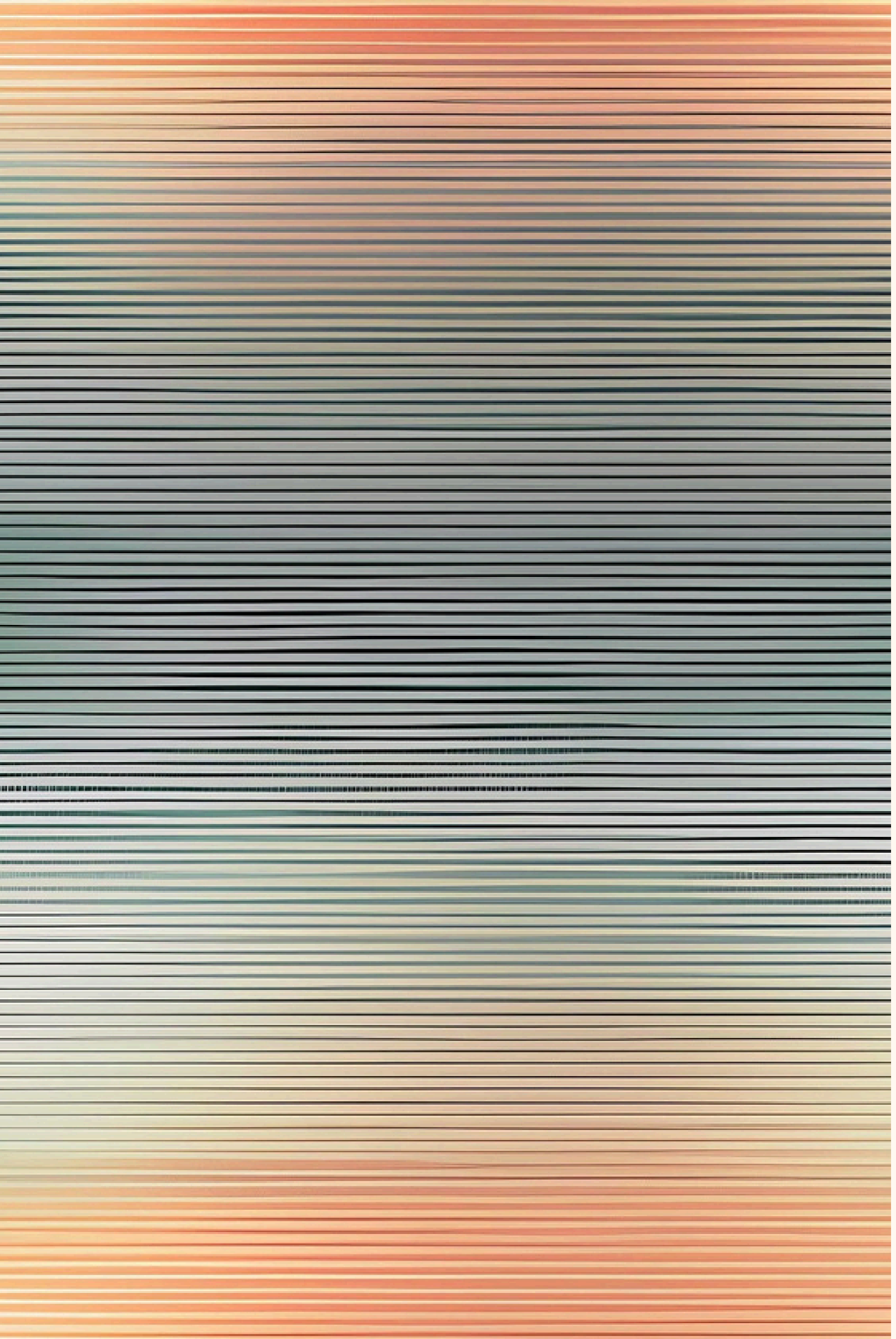
Problem Statement

What is the problem?

Some customers complete purchases (buyers) while others only browse. Reasons depend on engagement level and actions on the platform.

Main Goal

Analyse how engagement influences purchasing, identify differences between buyers and browsers, and determine actions predictive of purchase.



Dataset Overview

The dataset contains session-level interaction and behaviour data from an e-commerce platform, including engagement measures and purchase outcomes for each user session.

- Pages_Visited
- Time_Spent
- Items_Viewed
- Cart_Additions
- Previous_Purchases
- Device_Type
- Purchase (Yes / No)

Variables: Dependent & Predictors

Dependent

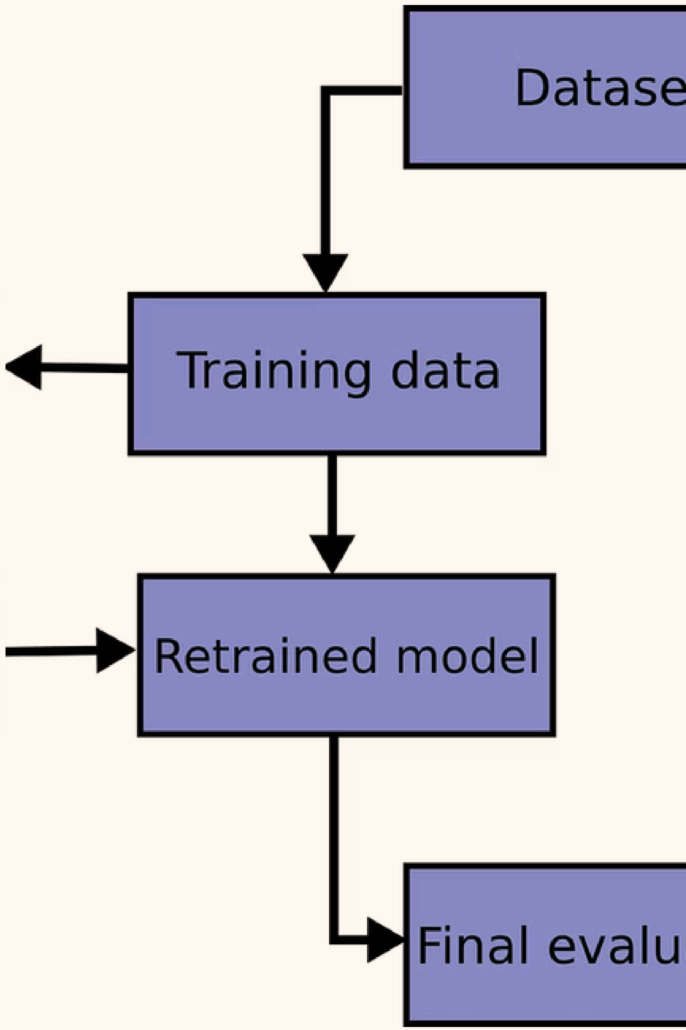
Purchase — Binary outcome (Yes = Buyer, No = Browser).

Predictors

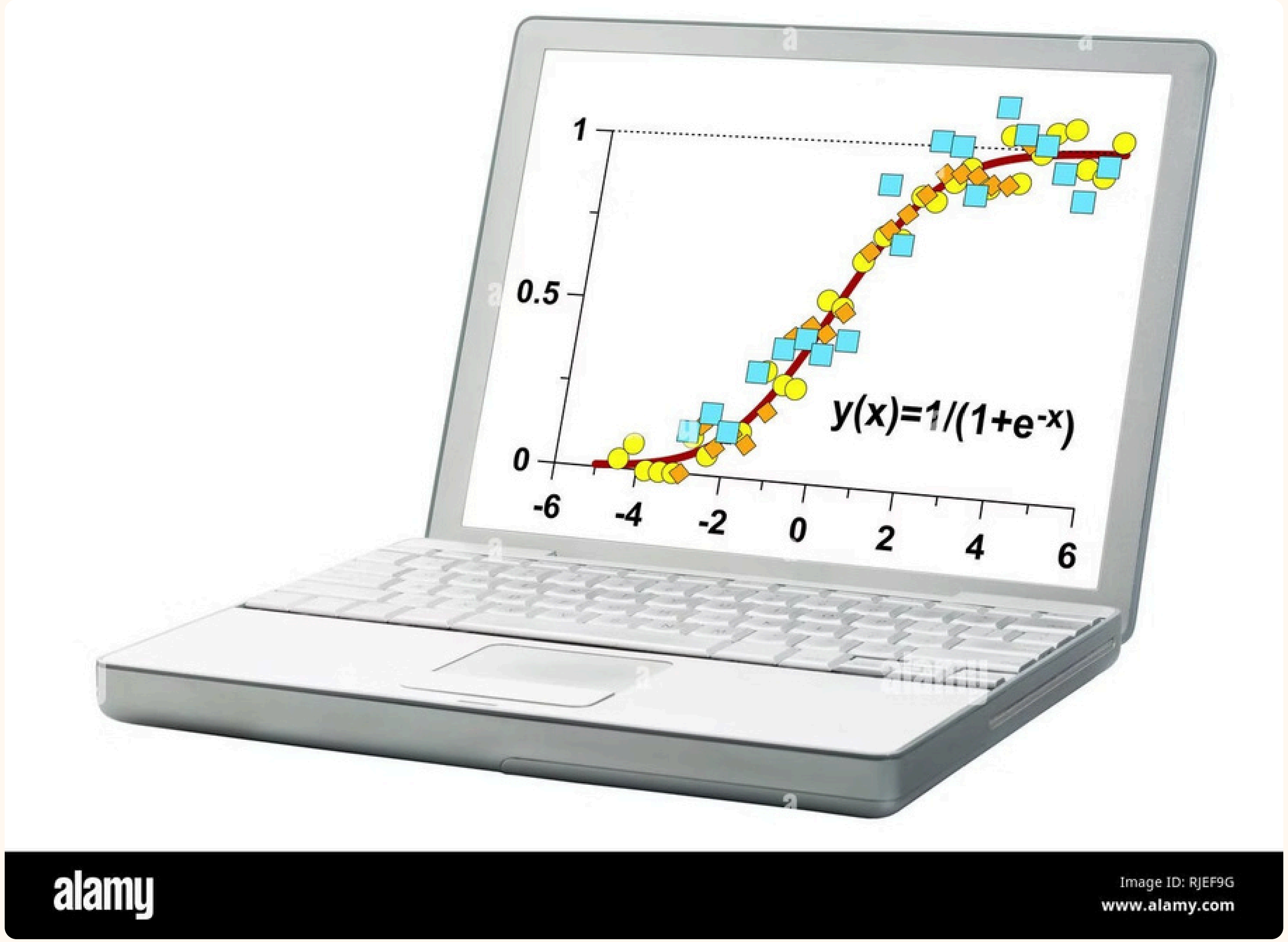
Pages_Visited, Time_Spent, Items_Viewed, Cart_Additions, Previous_Purchases, Device_Type.

Why Model Selection Matters

Choosing an inappropriate model can mislead conclusions. Model choice depends on dependent variable type (binary), predictor types(numerical & categorical) and research aims (prediction & group comparison).



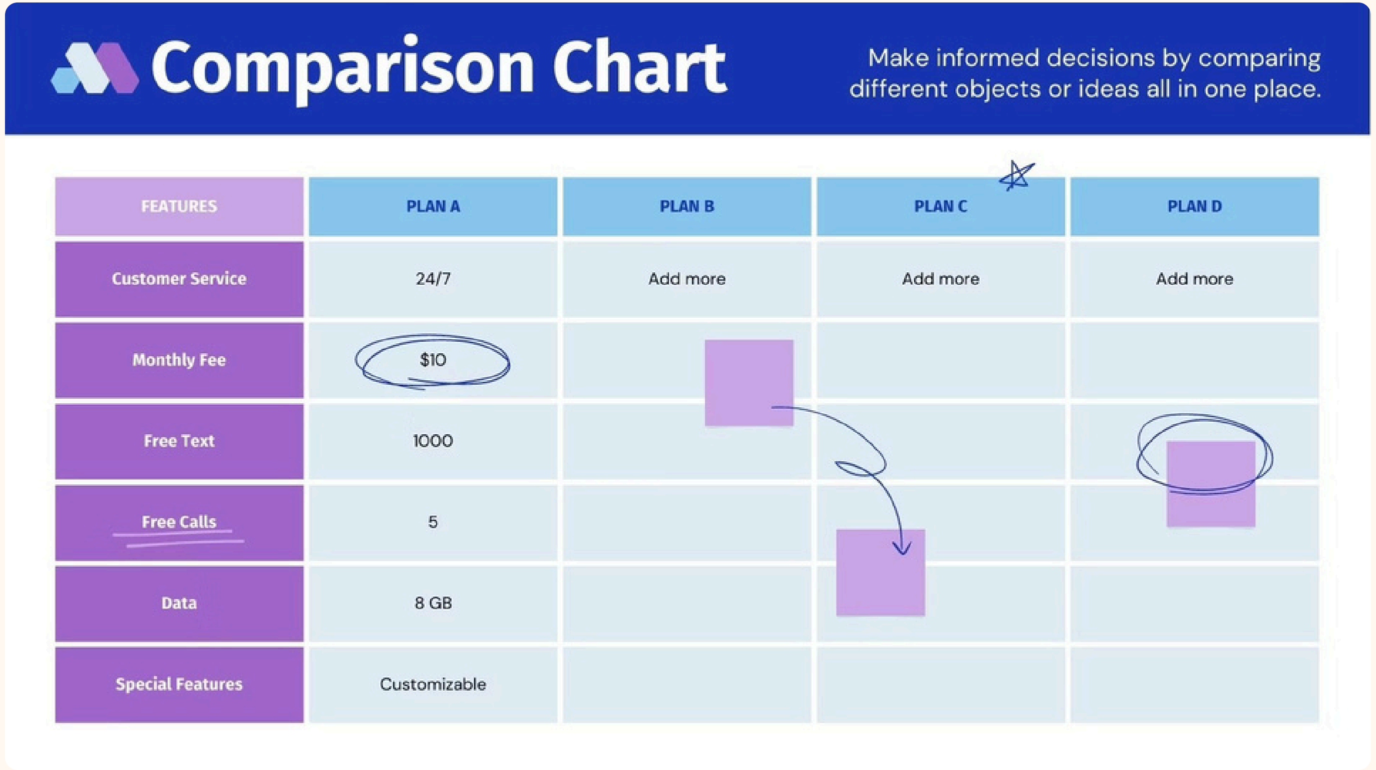
Logistic Regression — Primary Choice



Why?

Purchase is binary. Logistic regression estimates the probability of purchase based on engagement variables and identifies which actions significantly increase purchase likelihood.

ANOVA — Group Comparisons

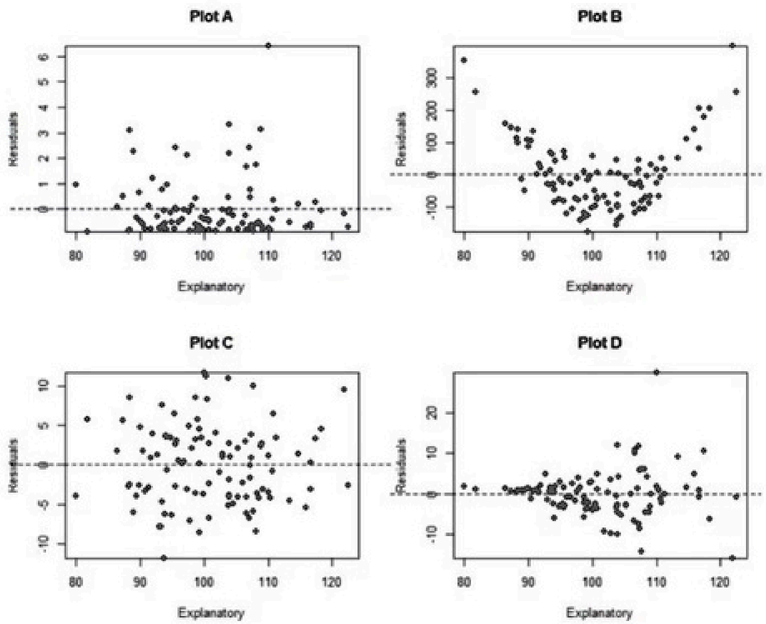


Purpose

Comparemean engagement (Time_Spent, Pages_Visited, Items_Viewed) across groups (buyers vs browsers or device types) to test for significant differences.

Multiple Linear Regression — Supporting Model

Which of the following plots shows a violation of the linearity assumption in regression?



- ◯ Plot A
- ◯ Plot B
- ◯ Plot C
- ◯ Plot D

Purpose

Explore how continuous engagement measures relate to each other. Support feature selection by identifying predictors that drive engagement intensity (e.g., Time_Spent explained by Pages_Visited).

Feature Selection

Cart_Additions

Direct indicator of purchase intent; strongly differentiates buyers from browsers.

Pages_Visited & Items_Viewed

Higher counts reflect exploration and intent; correlate with purchase likelihood.

Previous_Purchases

Signals loyalty and propensity to buy again; adds predictive power.

Justification: EDA showed meaningful differences between buyers and browsers; features align with business relevance.

Hypotheses & Conclusions

Hypothesis 1 (Logistic)

H0: Engagement variables do not affect purchase probability. H1: Higher engagement increases purchase probability.

Hypothesis 2 (Cart)

H0:Cart additions have no effect. H1: Adding to cart raises likelihood of purchase.

Hypothesis 3 (ANOVA)

H0:Mean time spent equal for buyers and browsers. H1: Buyers spend more time.

Hypothesis 4 (Regression)

H0:Pages visited donot influence engagement. H1: More pages visited → higher engagement.

Conclusion: Engagement (time, pages, items viewed, cart additions, prior purchases) strongly influences purchasing behaviour. Logistic regression, ANOVA and linear regression together provide robust insight and support reliable predictions.